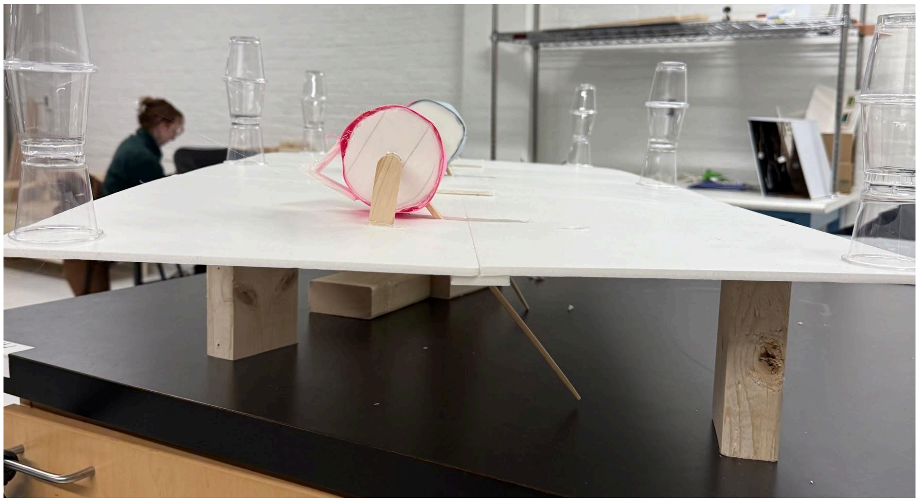
2.00C Sprint 4 Product - Team 2

User Interface Lever Prototype - MIT

We prototyped the surface with levers that the user would pull and interact with to cause the art on the wheel prototype to change. As the user pulls the lever, part of the wheel presses down on a button that actuates the mechanism.

This is a photo of the levers as seen from the top and side. They start off facing away from the user.



This is a view of the bottom of the mechanism; the white triangle presses down on the button as the lever is pulled.

Wheel Mechanism Prototype

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For this sprint, our team decided to create a prototype based on a wheel that shows different pieces of art. To achieve this, we created a model with a programmed motor that rotates a wheel by pressing a button. For this prototype, we used *Altered Landscapes* as the collection of choice.

Users can press one of three buttons, each featuring a symbol related to a specific painting. When a button is pressed, the wheel rotates to display the chosen artwork. This mechanism was built using *Arduino*, incorporating three buttons and a stepper motor. The buttons have three angular positions: 120°, 240°, and 360°. The purpose of our prototype is to display some of the pieces of art that present the state of Nevada's landscapes, captivating the audience, but at the same time, educating them on the importance of caring for their land. We feel that this could be achieved with this model and its special mechanism.

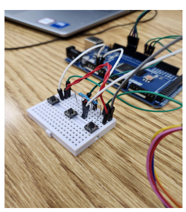

Figure 1: User buttons to move the wheel with one click

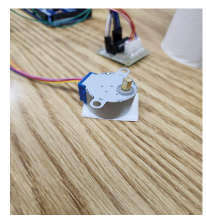

Figure 2: The stepper motor utilized to move the wheel

Figure 3: Three drawings representing artworks from *Altered Landscapes* Collection

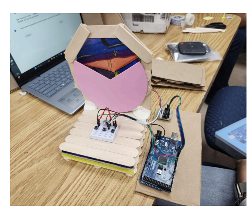

Figure 4: Finished prototype

Platform Prototype:

Jackie Frank

This model is an example of what the platform may look like, after completion it reminds me of an arcade style platform. The goal of this model is simplicity and for the viewer to have a basic understanding of the levers and buttons. This model has 3 buttons and 3 levers, each having its own mechanism for the picture being displayed. For example, the buttons can represent controls such as scent, color, and sound. The levers can represent a mine detonator, speed of wheel, and resistance. This is just a prototype, of course there is room for change and other ideas. The model has room for labeling the buttons and basic instruction and explanation of our model as a whole.

Figure 1:
Top view of the model, basic text will be included for guidance.

Figure 2:
Straight view, with the wheel in the back of it may look to the viewer.